# Read This First

**Delete this slide and every yellow hint box before you present.**

### 8 minutes or less
Practice out loud with a timer.

### Both partners speak
About equal time for each of you.

### Tell a story
Connect each slide to the next. Not just the rubric in order.

### Photos from YOUR experiment
No GIFs or random pictures.

### Label AI images
Write "AI-generated" under any AI-made image.

### Capitalize slide titles
Use Title Case on every slide.

### Tables and graphs
Same slide. UNITS, Capitalized Words, right graph type.

### No "human error"
Name problems in your design and how to fix them.

### Consistent APA
Same style on source slides and References.

### Few words, no reading
Short phrases. Do not read your slides.

**Check the rubric on the class site, then use the Grade Yourself page before you present.**

# Your Experiment Title Goes Here

Final Presentation

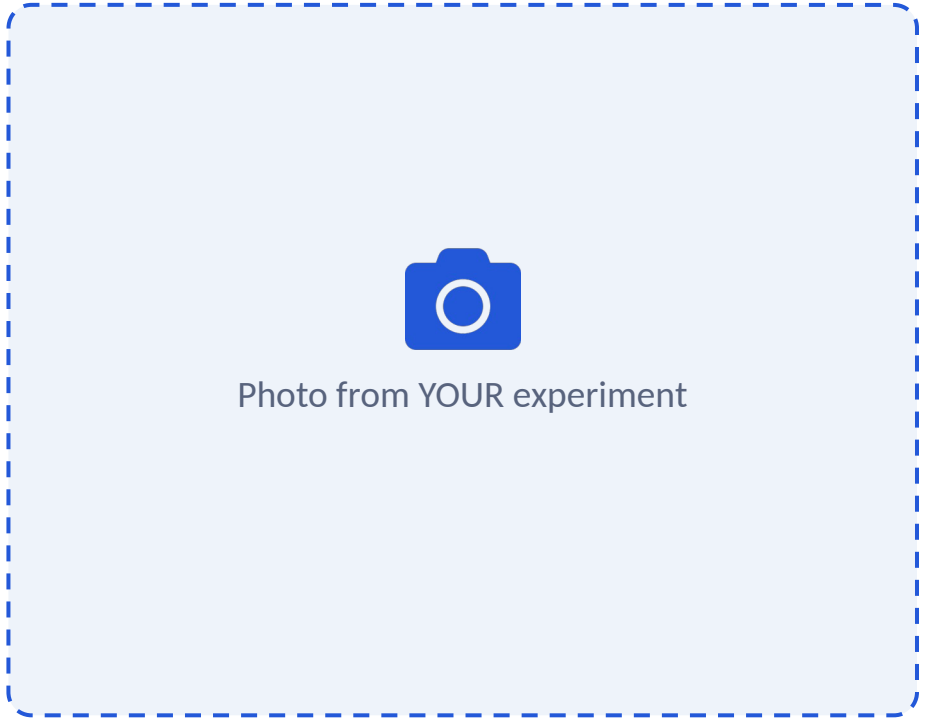
Partners: First Name and First Name



# Why This Matters

**Did you know _____?**

**We wanted to find out _____.**




**Hint (delete this box):** Open with a reason to care. Your talk is a story, not just the rubric in order. Connect each slide to the next.

# Background: Source 1

### APA Citation (Source 1)

Last, A. A., & Last, B. B. (Year). Title of the article in sentence case. Journal Name, volume(issue), pages. https://doi.org/...

### Summary

In Last name (year), researchers tested _____ on _____ and found _____.

### What This Means for Our Experiment

This tells us _____, so we expect _____.



Photo from YOUR experiment that connects to this source

**Hint (delete this box):** Use the citation builder on the class site. Match your References slide exactly.

**Hint (delete this box):** One source per slide. Every source needs a citation, a summary, and a "What This Means" statement.

# Background: Source 2

### APA Citation (Source 2)

Last, A. A., & Last, B. B. (Year). Title of the article in sentence case. Journal Name, volume(issue), pages. https://doi.org/...

### Summary

In Last name (year), researchers tested _____ on _____ and found _____.

### What This Means for Our Experiment

This tells us _____, so we expect _____.



Photo from YOUR experiment that connects to this source

**Hint (delete this box):** Use the citation builder on the class site. Match your References slide exactly.

**Hint (delete this box):** One source per slide. Every source needs a citation, a summary, and a "What This Means" statement.

# Our Question and Hypotheses

### Research Question

Does _____ change _____?

### Hypothesis (If, Then, Because)

If _____, then _____, because _____ (Author, Year).

### Null Hypothesis (No Difference)

There will be no difference in _____ between _____ and _____.



Photo from YOUR experiment

**Hint (delete this box):** Your "because" needs a source. The null hypothesis says what you change makes NO difference. Your p-value tests it.

# How We Ran the Experiment




**Our setup: _____**




**Condition A: _____**




**Condition B: _____**




**Measuring: _____**

**Hint (delete this box):** Four photos of YOUR experiment, with a short caption under each. Not pictures from a search.

# Recap of the Experiment

### Research Question

Does _____ change _____?

### Independent Variable

_____

A: _____

B: _____

### Dependent Variable

_____

(units: _____)

### Trials

_____ of each condition,

_____ in all

### Hypothesis and Null

If _____, then _____.

Null: no difference in _____.

### Controls

_____, _____, _____

**Hint (delete this box):** Put this slide right before your results so the audience remembers what you tested.

# Results: Data Table and Graph

| Trial | A: _____ (units) | B: _____ (units) |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| Average | | |



Paste your graph here.
Right graph type. Title Case title and axis titles with units.

**Hint (delete this box):** Table and graph go together on the SAME slide. Table: Capitalized headers, units, alternating colors, outlined cells, and Before, After, and % Change columns for heart rate or time. Graph: the right type for your data, Title Case titles, and units on the axes.

# Data Analysis and P-Value

| _____ | _____ | _____ (___%) |
| --- | --- | --- |
| Average of A | Average of B | Difference |

### Our P-Value

p = _____

Is it below 0.05? Yes / No

### What This Means (in Our Own Words)

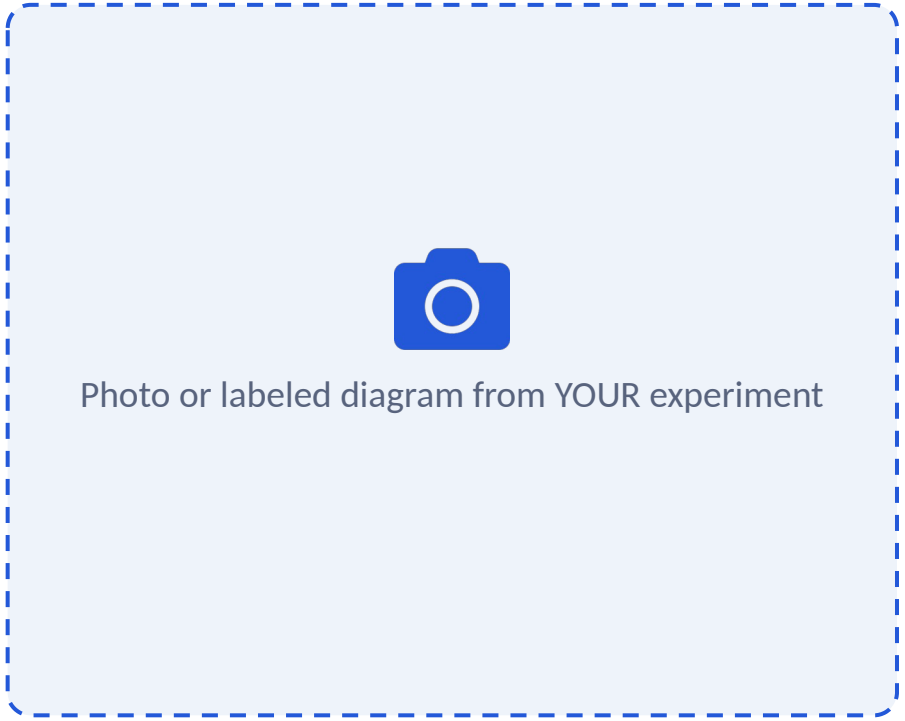
Because the p-value is _____, the difference is _____, and we _____ the null hypothesis.

**Hint (delete this box):** Use numbers to compare your conditions. Explain what the p-value means in your own words. Copy from the Conclusion Packet, Part A.

# Why It Happened

During exercise, the body _____, which causes _____.

Scientifically, this makes sense because _____.




**Hint (delete this box):** Use science words (for example: heart, oxygen, muscles, blood flow, breathing). Explain WHY your results happened. Copy from the Conclusion Packet, Part B.

# Conclusion: Was Our Hypothesis Supported?

**Claim**

Our hypothesis was _____ (supported / not supported).

**Evidence**

Our averages were _____ and _____. Our p-value was _____.

**Reasoning**

This happened because _____.

**Hint (delete this box):** Say clearly whether the hypothesis was supported, then back it up with your numbers and the p-value. Copy from the Conclusion Packet, Part C.

# Errors and How We Would Fix Them

**No "human error". Name a problem in the design, equipment, measuring, or uncontrolled variables.**

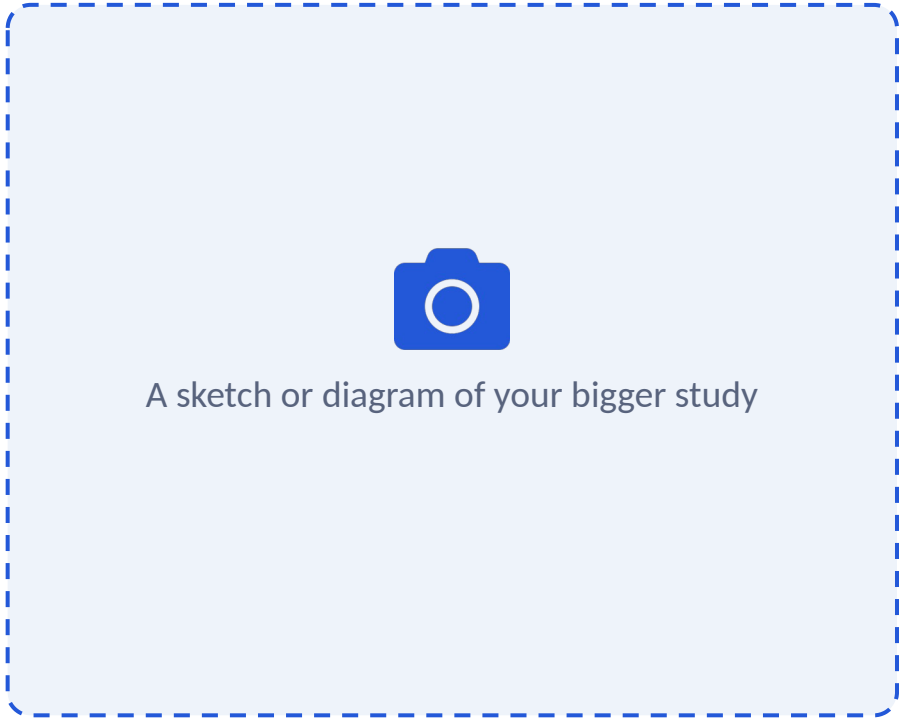
| | |
|---|---|
| **Error 1**<br>Because _____, our results may _____. | **How We Would Fix It**<br>Next time we would _____. |
| **Error 2**<br>Because _____, our results may _____. | **How We Would Fix It**<br>Next time we would _____. |
| **Error 3**<br>Because _____, our results may _____. | **How We Would Fix It**<br>Next time we would _____. |

**Hint (delete this box):** Be specific. "Be more careful" does not count. Change something about the experiment. Copy from the Conclusion Packet, Part D.

# Future Expansion

If we had millions of dollars and 5 years, we would expand this experiment by _____.

We would collect data from _____ and use _____ to learn _____.




**Hint (delete this box):** A creative, realistic bigger study with a clear purpose. If you use an AI image, label it "AI-generated". Copy from the Conclusion Packet, Part E.

# References

Last, A. A., & Last, B. B. (Year). Title of the article in sentence case. Journal Name, volume(issue), pages. https://doi.org/...

Last, A. A., & Last, B. B. (Year). Title of the article in sentence case. Journal Name, volume(issue), pages. https://doi.org/...

**Hint (delete this box):** Alphabetical by the first author's last name. Use the same APA style as your source slides. Italicize the journal name and the volume number.

# Questions from the Audience

Thank you!

